GUC
German University in Cairo
الجامعة الألمانية بالقاهرة

DMET 703: Video and Audio Technology

Lecture 7: Variable Length Encoding II

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Outline

- **Variable Length Coding**
  - **Huffman Coding**
  - **Arithmetic Coding**

Arithmetic coding

- Huffman coding:
  - Code word length is integer number of bits.
  - A code-word determines a symbol
- Arithmetic:
  - Replace the entire input with a single floating-point number, between 0 and 1.
  - Fractional code-word length
  - The code word itself defines an interval of real numbers between 0 and 1.

Arithmetic coding

- An entire sequence of source symbols (or message) is assigned a single arithmetic code-word.
  - **a one-to-one correspondence between source symbols and code words does not exist.**
- As the number of symbols in the message increases, the interval used to represent it becomes smaller and the number of information units (say, bits) becomes larger.
- Each symbol of the message reduces the size of the interval in accordance with the probability of occurrence.
- Decoding is not as fast as Huffman decoding, why?

Arithmetic coding

- The output of an arithmetic coding is a stream of bits
- the stream represents a fractional binary number between 0 and 1

$$01101010 \quad \rightarrow \quad 0.01101010$$

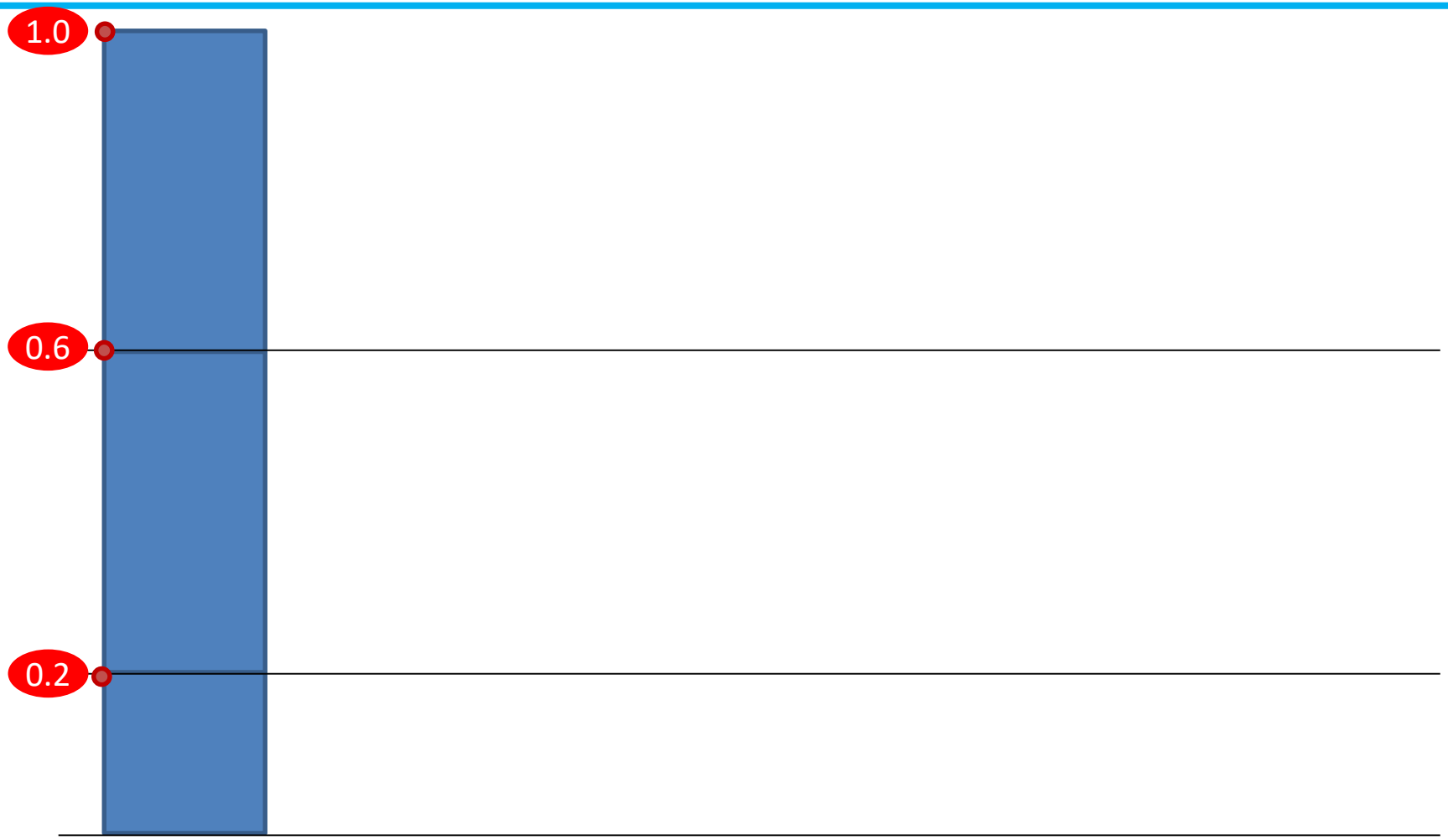
- Coding rate approaches high-order entropy theoretically. Working with finite precision (16 or 32 bits) causes compression be a little worser than entropy bound
- More complex than Huffman coding because multiplication and division operations are needed.

ARITHMETIC ENCODING

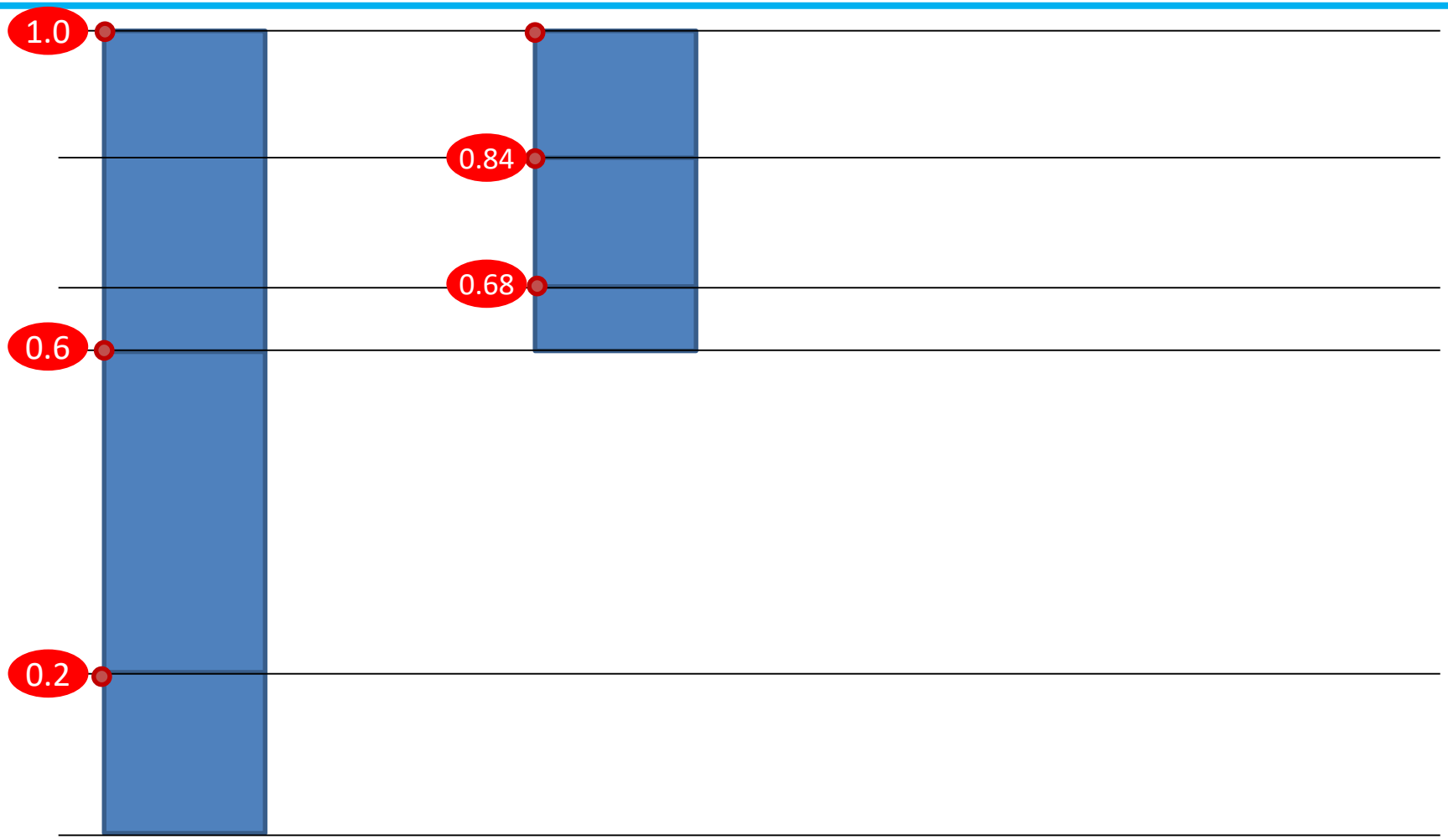
Arithmetic Encoding

- Source symbols: Xn = {a, b, c}
- P= (p0, p1, p2)= (0.2, 0.4, 0.4)
- Sequence = cba

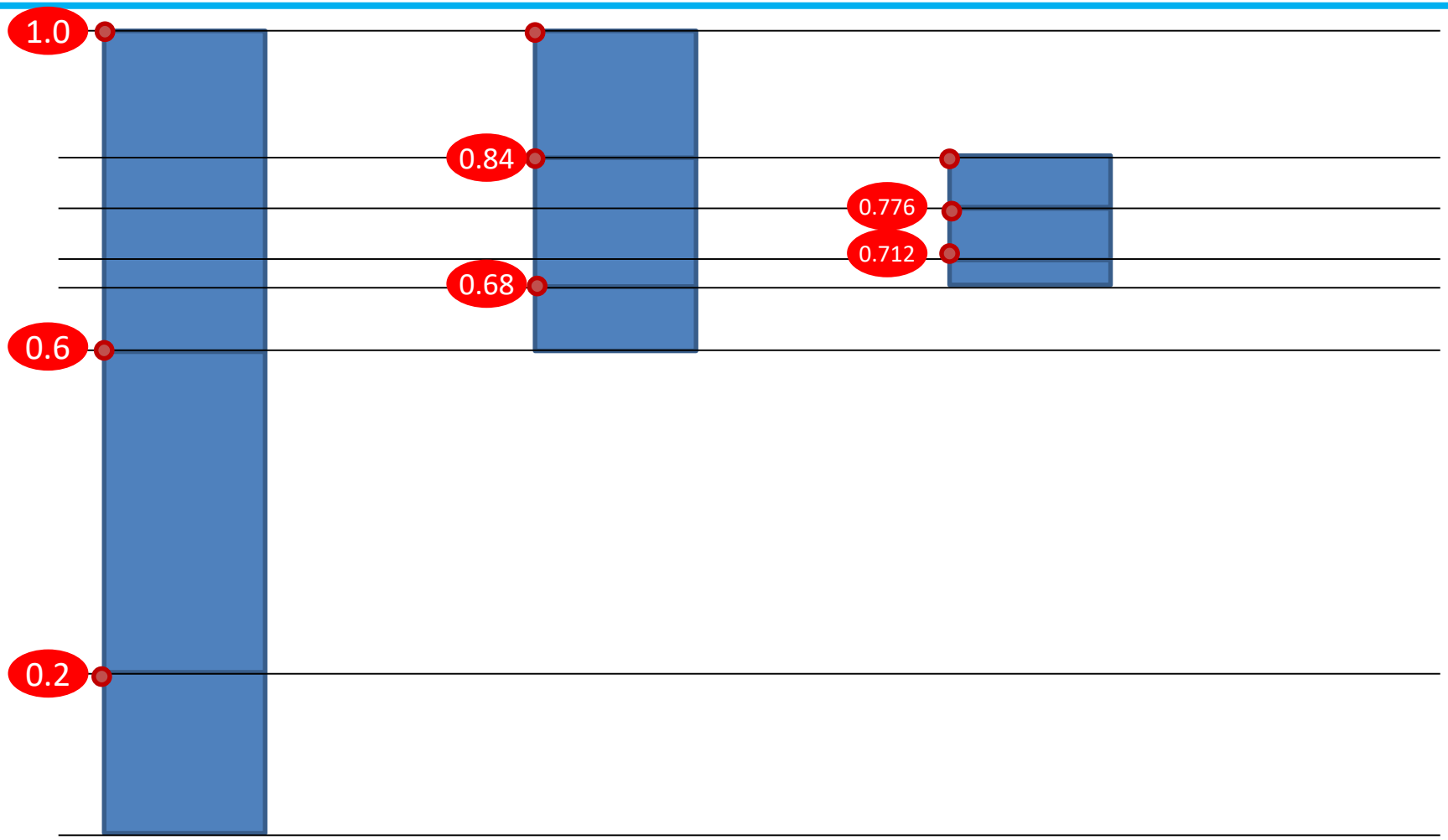
Arithmetic Coding



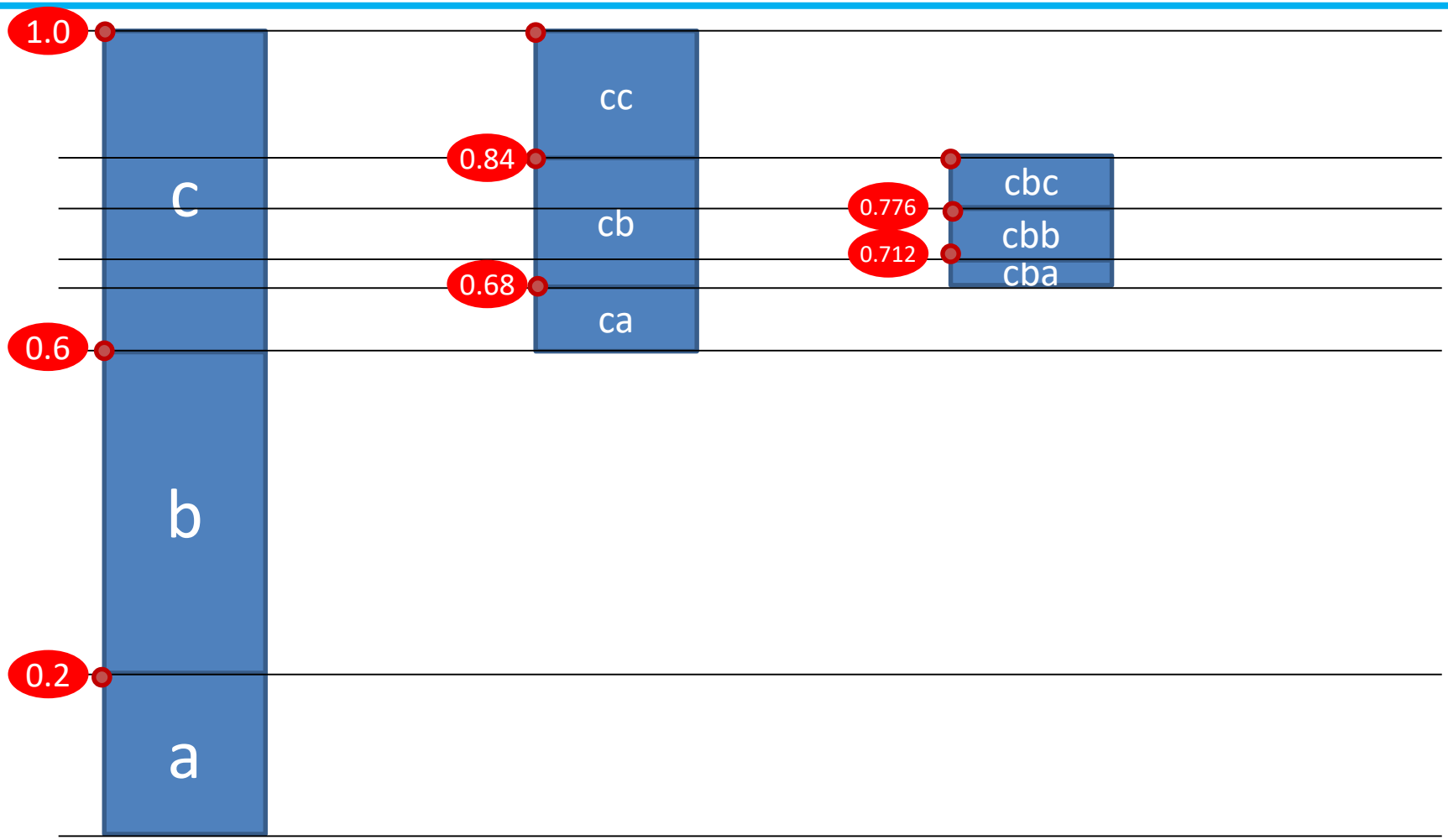
Arithmetic Coding



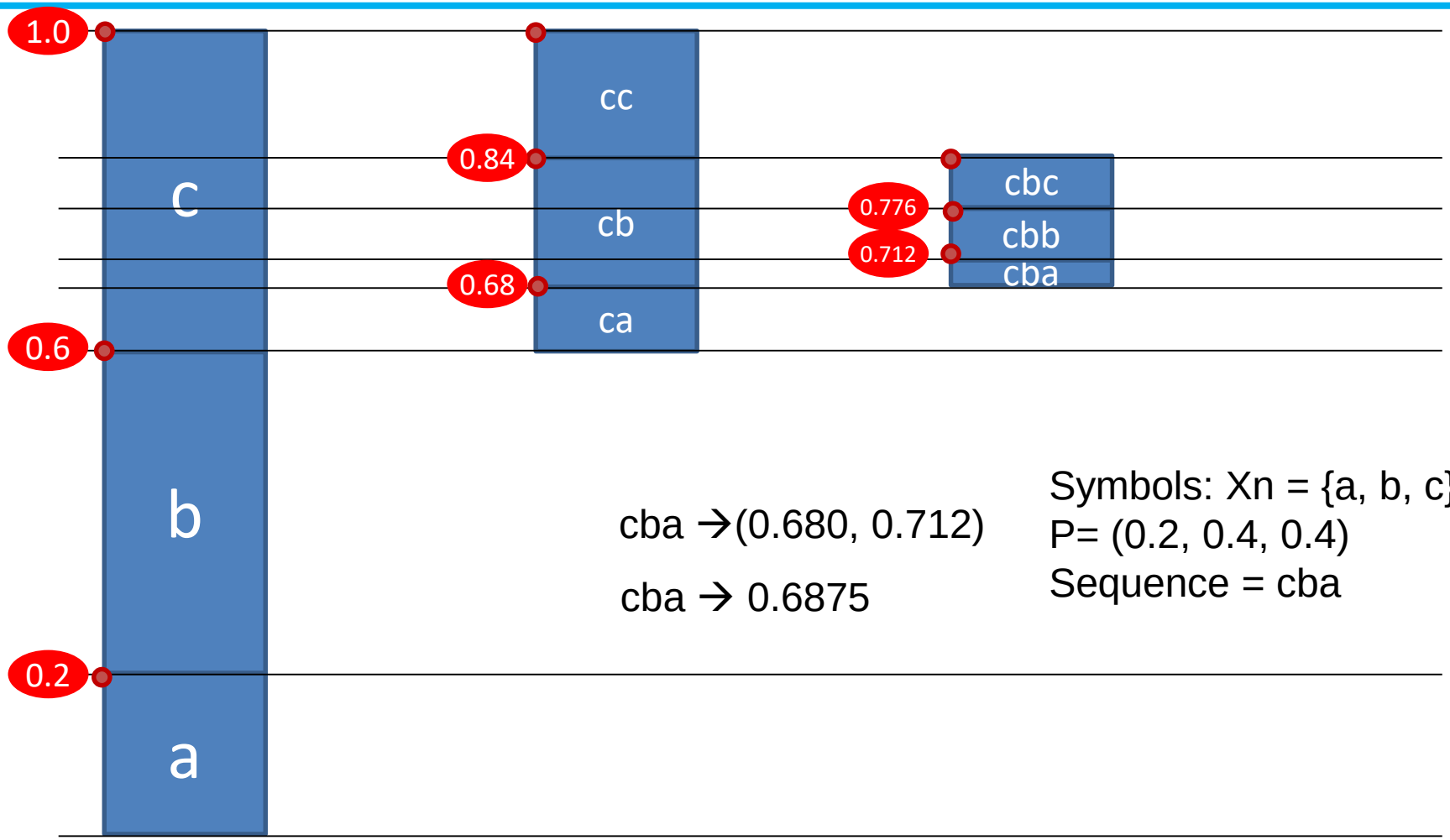
Arithmetic Coding



Arithmetic Coding



Arithmetic Coding



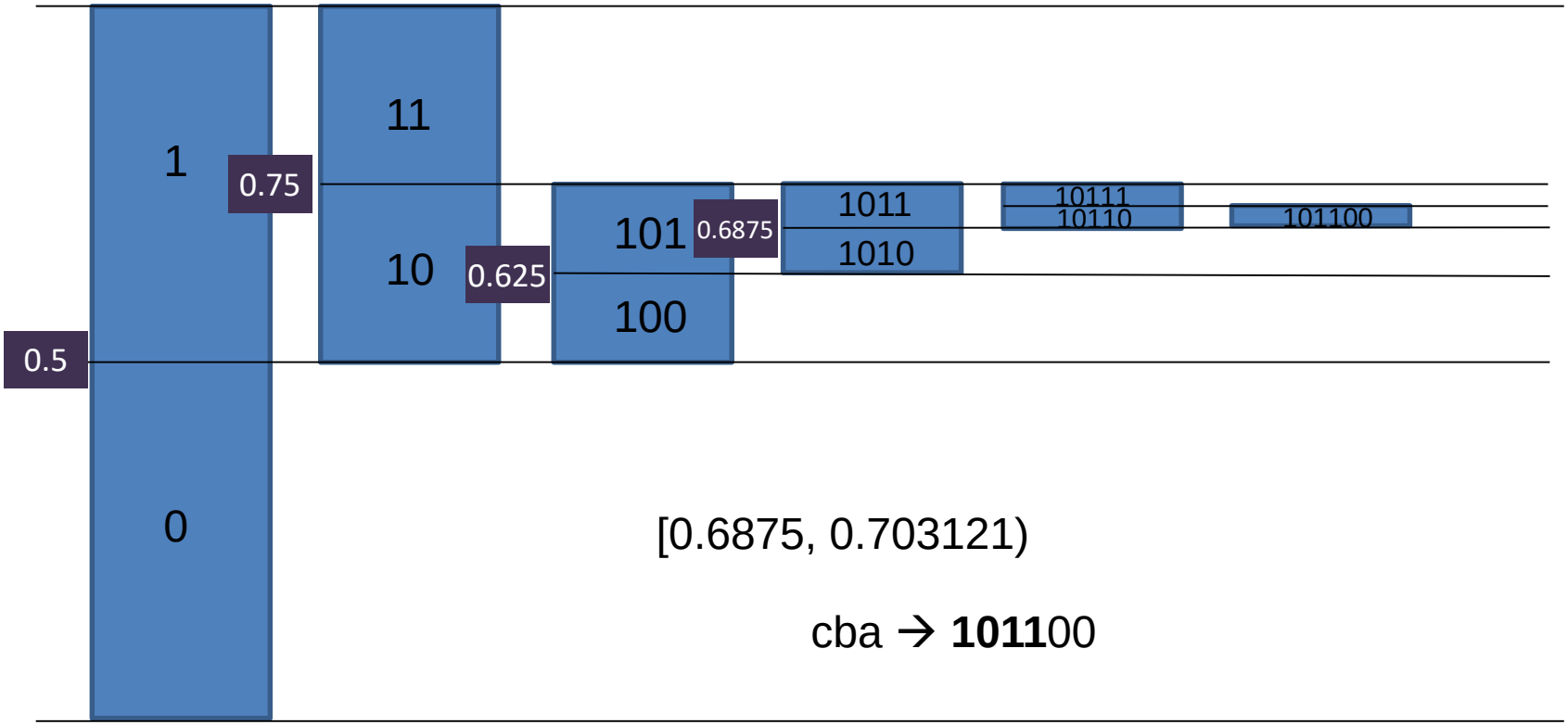
cba →(0.680, 0.712)

cba → 0.6875

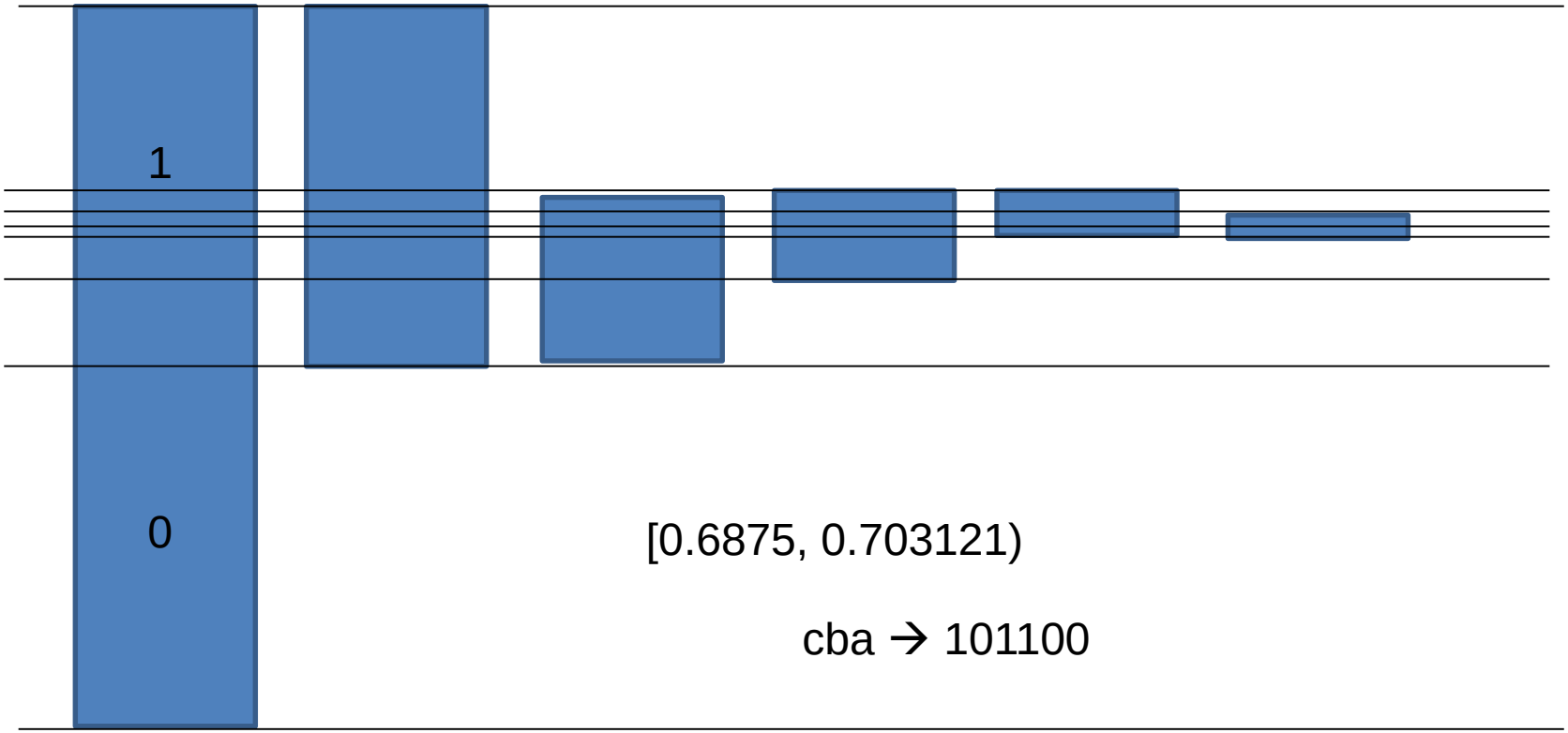
Symbols: Xn = {a, b, c}
P= (0.2, 0.4, 0.4)
Sequence = cba

Arithmetic Coding

Computing the binary representation

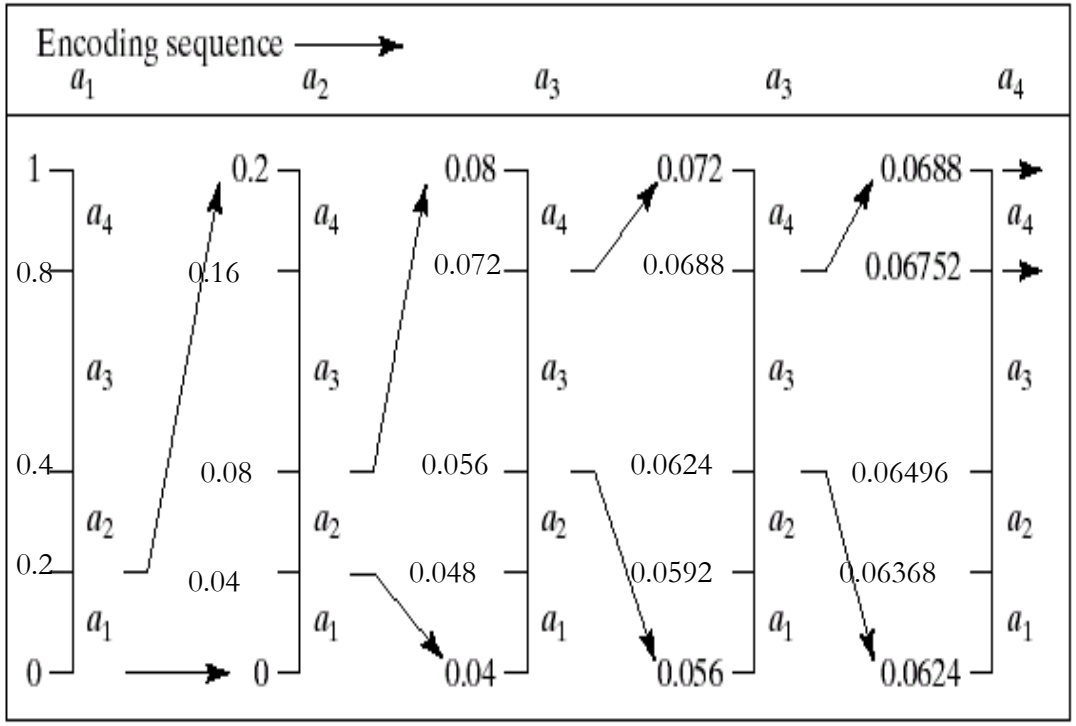


Arithmetic Coding



Source Symbol	Probability	Initial Subinterval
a_1	0.2	[0.0, 0.2)
a_2	0.2	[0.2, 0.4)
a_3	0.4	[0.4, 0.8)
a_4	0.2	[0.8, 1.0)

Let the message to be encoded be $a_1a_2a_3a_3a_4$


Encoding sequence
a_1
a_2
a_3
a_3
a_4
1
0.8
0.4
0.2
0
a_4
a_3
a_2
a_1
0.2
0.16
0.08
0.04
0
0.08
0.072
0.056
0.048
0.04
0.072
0.0688
0.0624
0.0592
0.056
0.0688
0.06752
0.06496
0.06368
0.0624

ARITHMETIC DECODING

Arithmetic Decoding

- Given the following set of symbols:

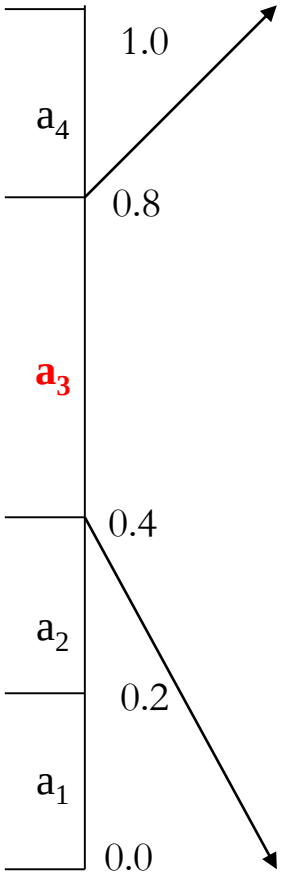
Source Symbol	Probability	Initial Subinterval
a_1	0.2	[0.0, 0.2)
a_2	0.2	[0.2, 0.4)
a_3	0.4	[0.4, 0.8)
a_4	0.2	[0.8, 1.0)

  - **Decode 0.572**

Arithmetic Decoding

Since 0.8>code word > 0.4, the first symbol should be a_3.

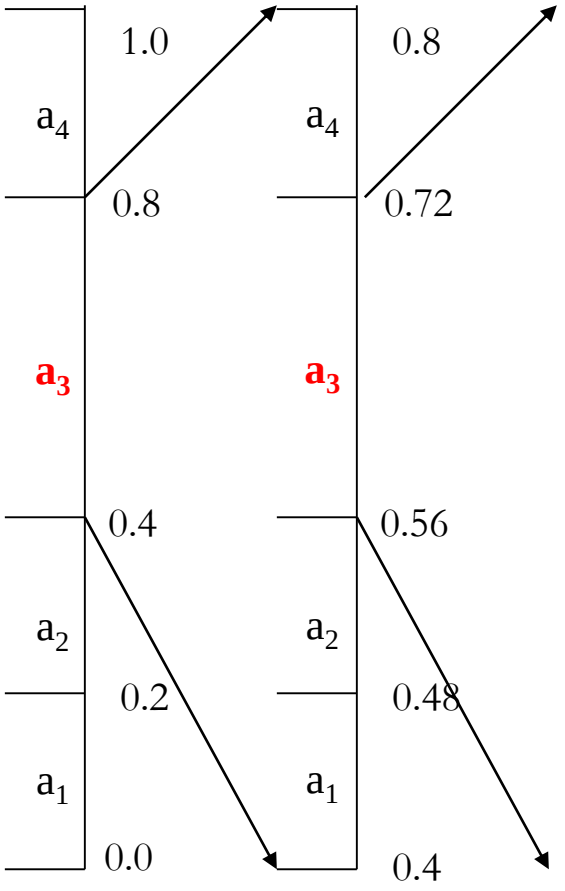
Decode 0.572.



Therefore, the message is : a_3

Arithmetic Decoding

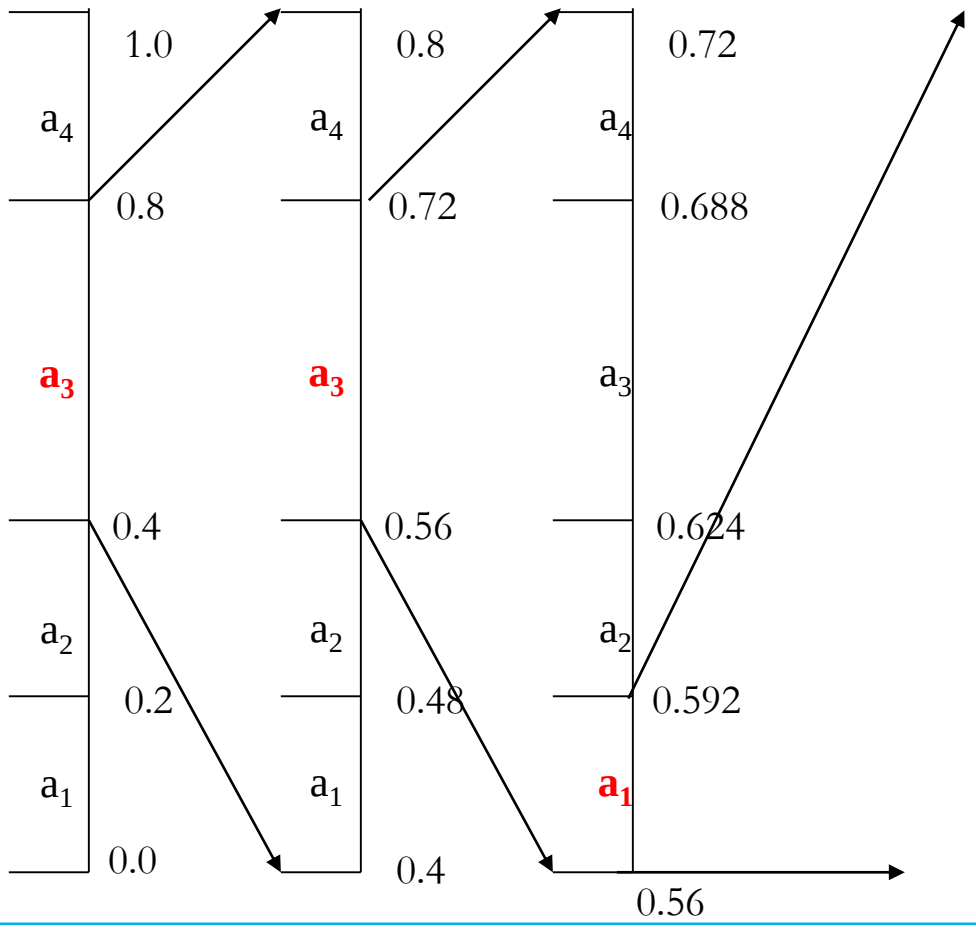
Since 0.8>code word > 0.4, the first symbol should be a_3.



Decode 0.572.

Therefore, the message is : a_3a_3

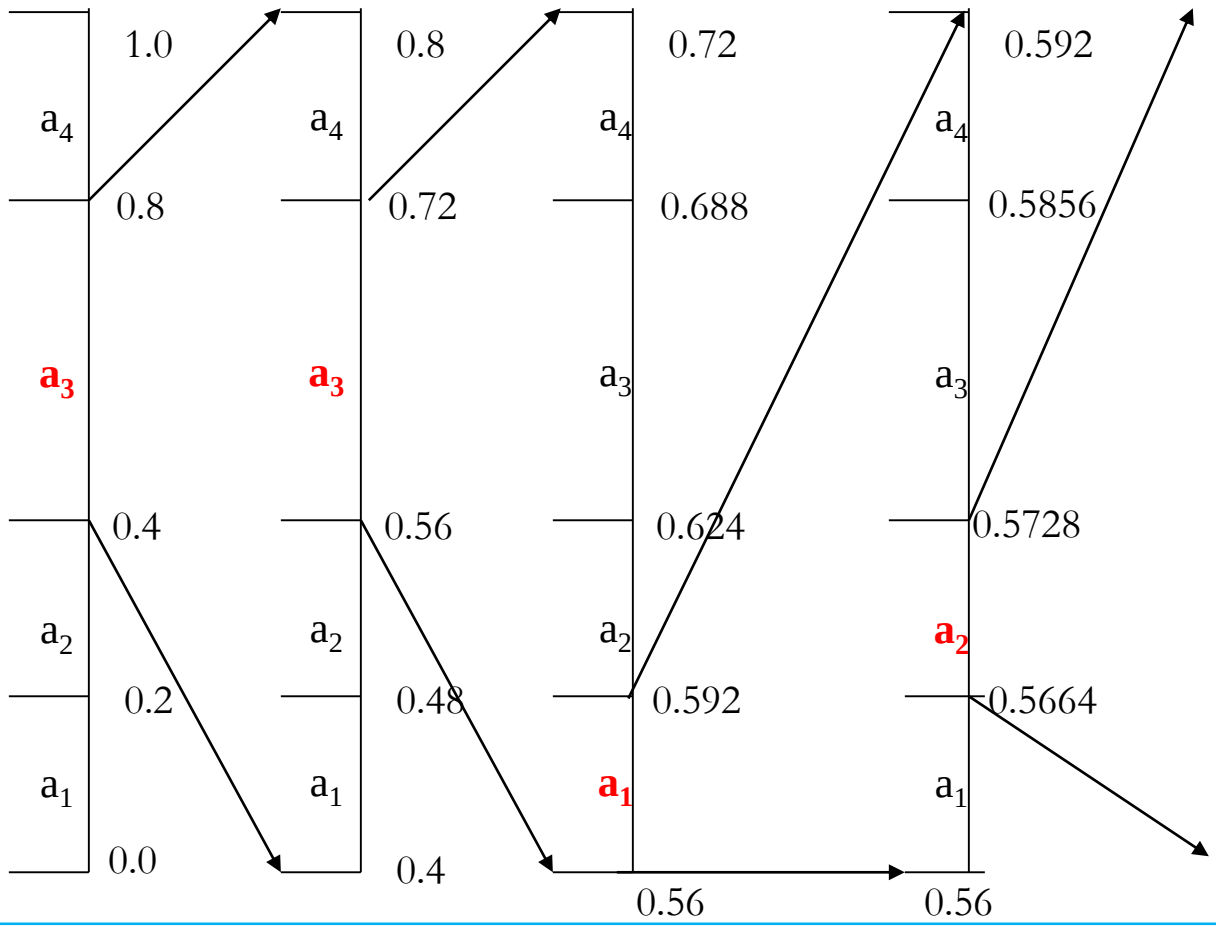
Arithmetic Decoding



Decode 0.572.

Therefore, the message is : $a_3a_3a_1$

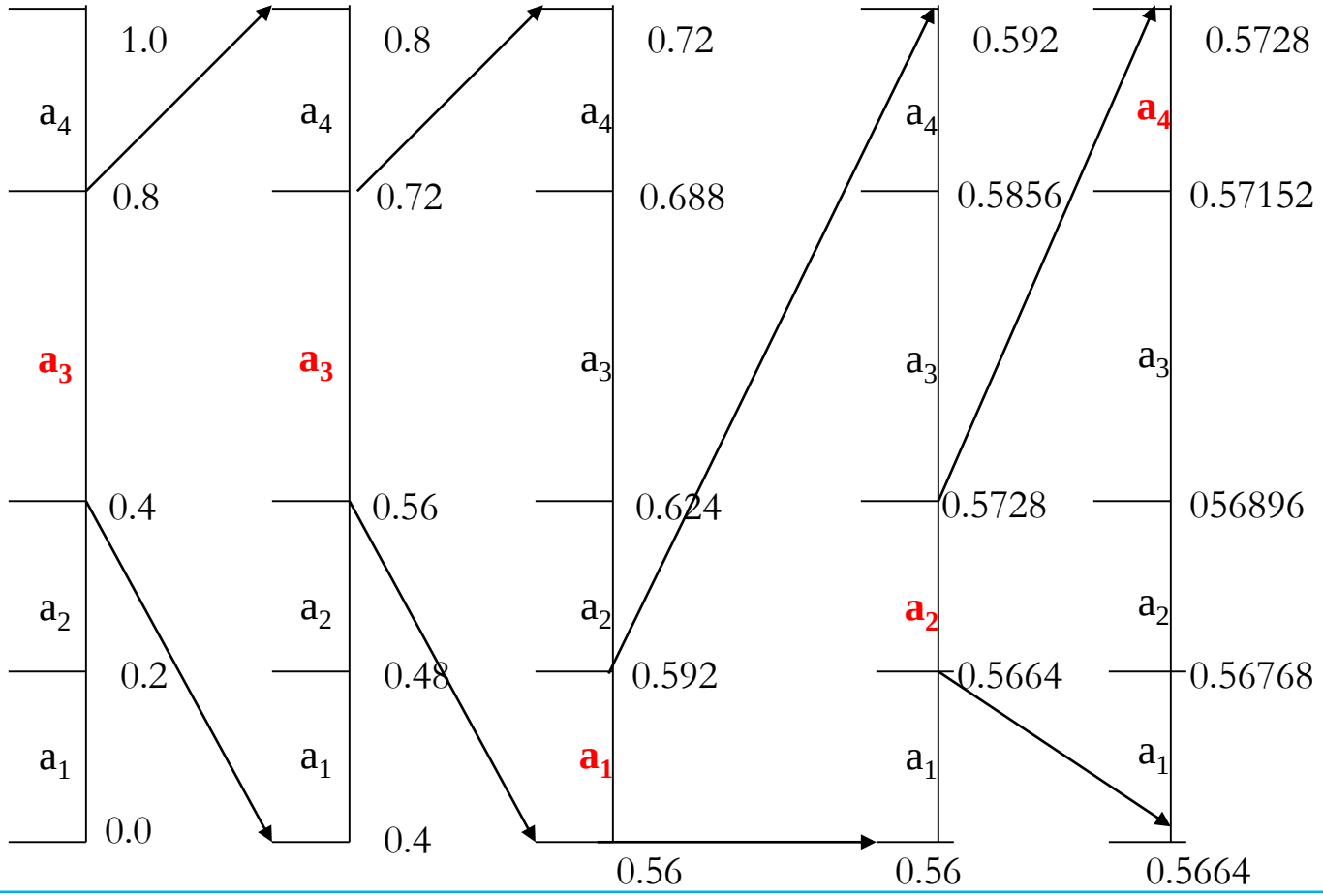
Arithmetic Decoding



Decode 0.572.

Therefore, the message is : $a_3a_3a_1a_2$

Arithmetic Decoding



Decode 0.572.

Therefore, the message is : $a_3a_3a_1a_2a_4$

Arithmetic Coding Summary

- In summary, the encoding process is simply one of narrowing the range of possible numbers with every new symbol.
- The new range is proportional to the predefined probability attached to that symbol.
- Decoding is the inverse procedure, in which the range is expanded in proportion to the probability of each symbol as it is extracted.
- Coding rate approaches high-order entropy theoretically.
- Not so popular as Huffman coding because $\times$, $\div$ are needed.

Any Questions?

